# Logic in Computer Science
## Problem Sheet 2: Proof Strategies
August 24, 2025

1. Suppose that $U_1$ and $U_2$ are nonempty sets and $U_1 \times U_2 \subseteq U_2 \times U_1$. Show that $U_1 = U_2$. Suggestion: show that $U_1 \subseteq U_2$ and $U_2 \subseteq U_1$, using proof by contradiction in each case.

2. Prove the following statement using contra-positive method:

   For all natural numbers p, if p is a prime number, then $\sqrt{p}$ is an irrational number.

3. Consider the following induction "proof" that all sheep in a flock are the same color:

   Base case: One sheep. It is clearly the same color as itself.

   Induction step: Let $n$ be an arbitrary natural number. Induction hypothesis: Any flock of $n$ sheeps are same color. Now consider A flock of $n+1$ sheeps. Take a sheep, a, out of the flock. The remaining $n$ are all the same color by induction hypothesis. Now put sheep a back in the flock, and take out a different sheep, b. By induction hypothesis, the $n$ sheep (now with a in their group) are all the same color. Therefore, a is the same color as all the other sheep; hence, all the sheep in the flock are the same color.

   What is wrong with this "proof"?

4. Suppose $A$ is a *wff* which doesn't use any negation symbol.
   (a) Show that the length of $A$ is odd.
   (b) Show that more than a quarter of the symbols are atomic proposition symbols.

5. Show that for every well formed formula $A$, every proper prefix $B$ of $A$, number of left parenthesis in B is less than number of right parenthesis in B.

6. Let c be a number of places at which binary connectives ($\wedge, \vee, \rightarrow, \leftrightarrow$) occur in a wff A and s be the number of places at which atomic proposition symbols occur in A. Prove that $s = c + 1$. For example, c and s are 6 and 7 respectively for the formula $(((p_1 \wedge p_2) \rightarrow p_3) \leftrightarrow ((p_1 \rightarrow p_2) \vee (p_1 \rightarrow p_3)))$.

7. A island is inhabited by two groups of people, the knights, who never lie and the knaves who always lie. A traveler comes across one the inhabitant and he says to the traveler, "This is not the first time I have said what I am now saying". What is the type of the inhabitant?